

Coding & Solution

Date	4 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	[Paste your GitHub / GitLab or hosted link here]
Programming Language(s)	Python, HTML, CSS
Framework(s) Used	Flask, Scikit-learn, XGBoost, Pandas, NumPy
Key Features Implemented	• Data preprocessing and feature engineering • Label encoding and feature scaling • Training multiple ML models (Logistic Regression, Decision Tree, Random Forest, XGBoost) • Automatic best model selection • Credit card approval prediction through a Flask web application • Performance evaluation and testing using Apache JMeter
Pending / Incomplete Features	Cloud deployment, Docker containerization, batch prediction, user authentication
Setup / Run Instructions	1. Install dependencies using <code>pip install -r requirements.txt</code>. 2. Navigate to the flask_app folder. 3. Run <code>python app.py</code>. 4. Open <code>http://127.0.0.1:5000</code> in a web browser. 5. Enter applicant details and click Predict to view the prediction result.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The **Credit Card Approval Prediction System** is implemented using Python and Flask with a modular project structure. The solution includes data preprocessing, feature engineering, machine learning model training, prediction, and evaluation modules. Multiple machine learning algorithms—Logistic Regression, Decision Tree, Random Forest, and XGBoost—were compared to select the best-performing model. The trained model is integrated into a Flask web application that predicts credit card approval based on user-provided applicant information. The project follows a clean folder structure, uses Git for version control, and includes comprehensive documentation for setup and execution.